

Assignment 3
CSE471: System Analysis and Design
Total Marks – 30

Q1. **[5]**

A software company is building an advanced application that deals with highly structured, interrelated entities and requires seamless interaction with an object-oriented programming environment. The system must preserve complex relationships and behaviors between data components, including inheritance and object identity. While flexibility in data modeling is essential, the development team prioritizes tight coupling between the application logic and the data model for performance and maintainability.

Which type of database is most suitable for this scenario and why?

Q2. **[10]**

Fields	Average size (Characters)
Order number	Your ID's first digit + 5
Date	Your ID's second digit + 5
Customer ID	Your ID's third digit + 5
Last Name	Your ID's fourth digit + 5
First Name	Your ID's fifth digit + 5
State	Your ID's sixth digit + 5
Amount	Your ID's seventh digit + 5
Tax rate	Your ID's last digit + 5

- a) Complete the table.
- b) Find the Record Size.
- c) Find the Total Record Size (Overhead = 0.35).
- d) Find Initial Table Volume (Initial Table Size = Summation of all the digits of your ID * 1200).
- e) Find the Final Table Volume after 1095 days (Quarterly growth rate = 1500).

Q3.

[5]

Choose five types of cohesion and give one example for each. (**Don't use slide examples.**)

Q4.

[10]

You are tasked with designing a Structure Chart for a simple Online Library Management System.

The system should support the following five main functions: 1. Registering a new user, 2. Displaying the catalog of books, 3. Searching for books, 4. Borrowing a book, 5. Returning a book. Each primary function should be divided into 2–3 sub-functions and follow a logical call order (e.g., users must register before accessing other features).

Details:

- **register_user()** should collect user input and validate it. If the input is invalid, the user is redirected to the homepage; otherwise, the user should proceed to create an account.
- **display_catalog()** should fetch all available books from the database, format the list, and render it to the user.
- **search_books()** should accept a user-provided search query, search the database, and display matched results.
- **borrow_book()** should check book availability, assign a due date, and update the system to mark the book as borrowed.
- **return_book()** should verify the book return, update the return status, check for overdue, and apply penalties if applicable.

Assume practical, logical data and control flows (couples) between the modules.